

Quiz 4 — Walking on a Graph: Solutions

M 375T Fall 2026 — 10 points. Q1: 5 pts; Q2: 5 pts.

1. (a) A 4-cycle: edges 0–1 (weight 2), 0–3 (weight 1), 1–2 (weight 1), 2–3 (weight 1); no edge 0–2 or 1–3. *2 pts: 1 for the four correct edges and no extras, 1 for the weights.* (b) $d = (3, 3, 2, 2)$ (row sums), and

$$P = \begin{pmatrix} 0 & \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

2 pts: 1 for d , 1 for P (rows must sum to 1; dividing by the wrong axis loses the point). (c) $\mu_1 = P^T \mu_0 = \text{row 0 of } P = (0, \frac{2}{3}, 0, \frac{1}{3})^T$. *1 pt.* [5 pts]

2. (a) $P^T \pi = \pi$ — equivalently $\pi^T P = \pi^T$, $(I - P^T)\pi = 0$, or $\sum_i \pi_i P_{ij} = \pi_j$ for every j , together with $\pi_i \geq 0$, $\sum_i \pi_i = 1$. *2 pts: the equation from memory is the point. $P\pi = \pi$ (wrong side) earns 1 pt if the computation in (c) is done consistently with row/column roles swapped correctly; stating “ $\pi \propto d$ ” alone is the answer to (b), not (a), and earns 0 here.* (b) $\pi = d / \sum_i d_i = (3, 3, 2, 2) / 10 = (0.3, 0.3, 0.2, 0.2)$ — by the lecture fact $\pi \propto d$ for an undirected walk, or by solving the linear system. *1 pt, normalization required.* (c) $\sum_i \pi_i P_{i0} = \pi_1 P_{10} + \pi_3 P_{30} = 0.3 \cdot \frac{2}{3} + 0.2 \cdot \frac{1}{2} = 0.2 + 0.1 = 0.3 = \pi_0$. *1 pt for the explicit arithmetic.* (d) π is the long-run fraction of time the walker spends at each vertex (equivalently, the law the walk settles into / the fixed point of the update). *1 pt, any sensible phrasing.* [5 pts]

Q1 mirrors Practice 5 A1–A3 (build d , P from W ; push μ one step) at one larger size, plus reading W from code; Q2 mirrors Practice 5 B7 with the defining equation deliberately not printed on the quiz — recalling $P^T \pi = \pi$ is the discriminator.